

## Midterm Test 2025. Row 2

1. Let  $A \subseteq \mathbb{R}$  be a bounded set. Prove that

1p

$$\forall \varepsilon > 0, \exists x \in A \text{ such that } x < \inf(A) + \varepsilon.$$

Proof. By definition,  $\inf(A) \leq x$ ,  $\forall x \in A$ , and if  $y \in \overline{\inf(A)}$ , then  $y \leq x$ ,  $\forall x \in A$ , then  $y \leq \inf(A)$ . (1) 0.5 p

Let  $\varepsilon > 0$ , then  $\inf(A) + \varepsilon > \inf(A) \xrightarrow{(1)} \inf(A) + \varepsilon \notin \overline{\inf(A)} \Rightarrow \exists x \in A \text{ s.t. } x < \inf(A) + \varepsilon$ . 0.5 p

2. Study the convergence of the following series:

4p

$$(a) \sum_{n=1}^{\infty} \frac{a^n}{a+n^2}, a > 0.$$

$$(c) \sum_{n=1}^{\infty} n(\sin \frac{1}{n} - \sin \frac{1}{n+1}).$$

$$(b) \sum_{n=1}^{\infty} \frac{2 \cdot 4 \cdot \dots \cdot (2n)}{1 \cdot 3 \cdot \dots \cdot (2n-1)}.$$

$$(d) \sum_{n=2}^{\infty} \frac{\ln n^2}{n^2}.$$

(a)  $x_n = \frac{a^n}{a+n^2}$ ,  $\frac{x_{n+1}}{x_n} = \frac{a^{n+1}}{a+(n+1)^2} \cdot \frac{a+n^2}{a^n} = a \cdot \frac{n^2+a}{(n+1)^2+a} \rightarrow a$  Ratio Test. if  $a < 1$ , convergent  
if  $a > 1$ , divergent. 0.75 p

For  $a=1$ ,  $x_n = \frac{1}{1+n^2}$ ,  $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$  is convergent as  $\sum_{n=1}^{\infty} \frac{1}{n^2} > 1$ . 0.25 p

(b)  $x_n = \frac{2 \cdot 4 \cdot \dots \cdot (2n)}{1 \cdot 3 \cdot \dots \cdot (2n-1)}$ ,  $\frac{x_{n+1}}{x_n} = \frac{2n+2}{2n+1} \rightarrow 1$  (Ratio Test is inconclusive) 0.25 p

Consider  $n \left( \frac{x_n}{x_{n+1}} - 1 \right) = n \left( \frac{2n+1}{2n+2} - 1 \right) = n \cdot \frac{-1}{2n+2} = -\frac{n}{2n+2} \rightarrow -\frac{1}{2} < 1$  Raabe-Duhamel: divergent. 0.25 p

(c)  $\sin \frac{1}{n} - \sin \frac{1}{n+1} = 2 \sin \frac{1}{2} \left( \frac{1}{n} - \frac{1}{n+1} \right) \cdot \cos \frac{1}{2} \left( \frac{1}{n} + \frac{1}{n+1} \right) = 2 \sin \frac{1}{2n(n+1)} \cdot \cos \frac{2n+1}{2n(n+1)} \rightarrow \cos 0 = 1$  0.25 p

Recall that  $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ , thus  $\lim_{n \rightarrow \infty} \frac{\sin \frac{1}{2n(n+1)}}{\frac{1}{2n(n+1)}} = 1 \Rightarrow \lim_{n \rightarrow \infty} n^2 \left( \sin \frac{1}{n} - \sin \frac{1}{n+1} \right) = \lim_{n \rightarrow \infty} n^2 \cdot \frac{2}{2n(n+1)} = 1$

$\Rightarrow 1 = \lim_{n \rightarrow \infty} n^2 \left( \sin \frac{1}{n} - \sin \frac{1}{n+1} \right) = \lim_{n \rightarrow \infty} n \cdot n \left( \sin \frac{1}{n} - \sin \frac{1}{n+1} \right) = \lim_{n \rightarrow \infty} \frac{n \left( \sin \frac{1}{n} - \sin \frac{1}{n+1} \right)}{\frac{1}{n}} = 1$  0.5 p

$\Rightarrow \sum_{n=1}^{\infty} n \left( \sin \frac{1}{n} - \sin \frac{1}{n+1} \right)$  has the same nature as  $\sum_{n=1}^{\infty} \frac{1}{n}$ : divergent. 0.25 p

Method 2: By the Mean-Value Theorem  $\sin \frac{1}{n} - \sin \frac{1}{n+1} = \left( \frac{1}{n} - \frac{1}{n+1} \right) \cos \frac{1}{\xi_n}$ ,  $\xi_n \in (n, n+1)$   
 $\Rightarrow n \left( \sin \frac{1}{n} - \sin \frac{1}{n+1} \right) = n \cdot \frac{1}{n(n+1)} \cos \frac{1}{\xi_n} \approx \frac{1}{n+1} \rightarrow$  divergent series.

Method 3:  $\sum_{n=1}^N n \left( \sin \frac{1}{n} - \sin \frac{1}{n+1} \right) = \sum_{n=1}^N n \sin \frac{1}{n} - \sum_{n=1}^N (n+1) \sin \frac{1}{n+1} + \sum_{n=1}^N \sin \frac{1}{n+1} = \sin 1 - (N+1) \sin \frac{1}{N+1} + \sum_{n=1}^N \sin \frac{1}{n+1}$   
 $\Rightarrow \sum_{n=1}^{\infty} n \left( \sin \frac{1}{n} - \sin \frac{1}{n+1} \right) = \sin 1 - \lim_{N \rightarrow \infty} (N+1) \sin \frac{1}{N+1} + \sum_{n=1}^{\infty} \sin \frac{1}{n+1}$  "like"  $\sum_{n=1}^{\infty} \frac{1}{n+1} = \infty$  divergent

(d)  $\sum_{n=2}^{\infty} \frac{\ln n^2}{n^2} = 2 \sum_{n=2}^{\infty} \frac{\ln n}{n^2}$  has the same nature as  $\int_2^{\infty} \frac{\ln x}{x^2} dx = -\frac{\ln x}{x} \Big|_2^{\infty} + \int_2^{\infty} \frac{1}{x^2} dx$  0.5 p

$= -\lim_{x \rightarrow \infty} \frac{\ln x}{x} + \frac{\ln 2}{2} - \frac{1}{x} \Big|_2^{\infty} = 0 + \frac{\ln 2}{2} - 0 + \frac{1}{2} = \frac{\ln 2 + 1}{2} \Rightarrow$  convergent series. 0.5 p

Method 2: Cauchy Condensation Test. same nature as  $\sum_{n=2}^{\infty} 2^n \frac{\ln 2^{2n}}{2^{2n}} = \sum_{n=2}^{\infty} \frac{2n \ln 2}{2^n}$  convergent by Ratio Test. 0.5 p

3. Compute  $\int_0^{\infty} x^3 e^{-x^2} dx = I$

1p

Let  $u = x^2$ ,  $du = 2x dx$ ,  $I = \int_0^{\infty} x^2 e^{-x^2} x dx = \int_0^{\infty} \underbrace{u e^{-u}}_{= (e^{-u})'} \frac{1}{2} du = -\frac{1}{2} u e^{-u} \Big|_0^{\infty} + \frac{1}{2} \int_0^{\infty} e^{-u} du$  0.25 p

Let  $u = x^2$ ,  $du = 2x dx$ ,  $\int_0^1 x^2 e^{-x^2} x dx = \int_0^1 \underbrace{u}_{=e^{-u}} e^{-u} \frac{1}{2} du = -\frac{1}{2} u e^{-u} \Big|_0^1 + \frac{1}{2} \int_0^1 e^{-u} du$  0.25p

$= -\frac{1}{2} \lim_{u \rightarrow \infty} \frac{u}{e^u} + 0 - \frac{1}{2} e^{-u} \Big|_0^1 = -\frac{1}{2} \frac{e^{-1}}{1} + \frac{1}{2} = \frac{1}{2}$  0.25p

4. (a) Find the Taylor series around 0 for  $\cos^2 x$ .

1p

(b) Find the higher order derivatives  $f^{(n)}(0)$ ,  $n \in \mathbb{N}$ , for the function

0.5p

$$f(x) = \begin{cases} \frac{\sin x}{x}, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0. \end{cases}$$

(a)  $\cos^2 x = \frac{1 + \cos 2x}{2}$ . We use the Taylor series for  $\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$  and obtain

$$\cos(2x) = \sum_{n=0}^{\infty} \frac{(-1)^n (2x)^{2n}}{(2n)!} = 1 + \sum_{n=1}^{\infty} \frac{(-1)^n 2^{2n} x^{2n}}{(2n)!} \Rightarrow \cos^2 x = 1 + \sum_{n=1}^{\infty} \frac{(-1)^n 2^{2n-1} x^{2n}}{(2n)!}$$
 0.25p 0.5p

(b) Taylor series formula:  $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n \Rightarrow$  to find  $f^{(n)}(0)$  we can use the Taylor series.

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} \Rightarrow \frac{\sin x}{x} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n+1)!} \Rightarrow \left[ f^{(2n+1)}(0) = 0, f^{(2n)}(0) = \frac{(-1)^n}{2n+1} \right]$$
 0.25p 0.25p

5. Find the radius of convergence and the sum of the power series

1.5p

$$\sum_{n=1}^{\infty} (-1)^n n^2 x^n.$$

$\bullet L = \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} (n+1)^2}{(-1)^n n^2} \right| = \lim_{n \rightarrow \infty} \left( \frac{n+1}{n} \right)^2 = 1 \Rightarrow R = 1$  (Radius of convergence): abs. conv.  $\forall x \in (-1, 1)$  0.25p

$\bullet$  For  $x \in (-1, 1)$ ,  $\sum_{n=0}^{\infty} (-1)^n x^n = \sum_{n=0}^{\infty} (-x)^n = \frac{1}{1-(-x)} = \frac{1}{1+x}$ .

Differentiating  $\Rightarrow \sum_{n=1}^{\infty} (-1)^n n x^{n-1} = -\frac{1}{(1+x)^2} \xrightarrow{\cdot x} \sum_{n=1}^{\infty} (-1)^n n x^n = -\frac{x}{(1+x)^2}$  0.5p

Differentiating  $\Rightarrow \sum_{n=1}^{\infty} (-1)^n n^2 x^{n-1} = -\frac{(1+x)^2 - x \cdot 2(1+x)}{(1+x)^4} = -\frac{(1+x)(1+x-2x)}{(1+x)^4} = -\frac{1-x}{(1+x)^3} \cdot x$

$\Rightarrow \sum_{n=1}^{\infty} (-1)^n n^2 x^n = \frac{x(x-1)}{(1+x)^3}$  0.5p